AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M269
Algorithms, Data Structures and Computability

Midterm Examination (MTA)
Spring Semester 2024/2025

Date: 8 /April/2025

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 2 hours

Instructions:

- Total Marks: **60**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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PART 1: ALL QUESTIONS ARE REQUIRED [10 Marks]

Question 1: (10 marks)

True or false

1) The Stack is a non-linear type of data structure
2) Finding the smallest element in a linked list of size M is of O (log M).
3) In a queue, If we implement enqueue(7), enqueue(2), dequeue(), enqueue(2), enqueue(7), dequeue(), dequeue(), and enqueue(7) the remaining value in the queue will be 7.
4) The selection-sort performs equal swapping steps in comparison with the bubble-sort algorithm.
5) 1000n is less in complexity than $2n^2$
6) Heap sort algorithm sorts use auxiliary memory more than the merge sort
7) When determining the efficiency of an algorithm, the space factor is measured by counting the average memory needed by the algorithm.
8) The queue is best data-structure in case the given data is served first in last out.
9) If $f(n) = \theta\, g(n)$ then $g(n) = \theta\, f(n)$
10) the following code is used to remove current node in a double linked list:

```
current.prev.next = current.next;
current.next.prev = current.prev;
current = current.prev;
```

PART 2: ALL QUESTIONS ARE REQUIRED [20 Marks]

Question 2: (8 marks)

a) Discuss briefly the difference between the Computational Thinking and the Computational Problem. (2 marks)
b) Mention two limitations of conducting experimental analysis to study the complexity of a given algorithm. (2 marks)
c) Explain the order of elements inside the Stack data structure versus the Queue Data structure (2 marks)
d) Which is a less complexity: (3n) or (100 lg n)? Justify your answer. (2 marks)

Question 3: (12 marks)

Given a group of N books of different sizes are placed on a table. You are asked to rearrange the books as illustrated in the Fig.1. (Total 12 marks, 2 marks each)

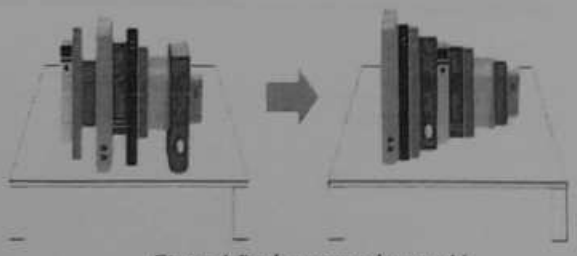


Figure 1 Books arranged on a table

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a) If the only thing you are allowed to do is to swap any two adjacent book
   1) This problem is similar to a famous sorting algorithm. What is the name of this algorithm?
   2) What is its complexity?

b) If the only thing you are allowed to do is to pull out one book from anywhere and to place it again at the left shifting the entire package to the right.
   3) This technique is almost similar to a famous sorting algorithm. What is the name of this algorithm?
   4) Given that you have quick eyes, you can immediately determine which book is the smallest or largest to pull it out with O(1), what is its complexity of the overall arranging technique.

c) If the only thing you are allowed to do is to start the rearrangement of the books by considering book by book starting from the right. Every book is compared with the stack of books on the right, then pulled from its position to be placed in the correct position to keep the stack of books on the right correctly arranged till all the books are arranged.
   5) This technique is almost similar to a famous sorting algorithm. What is the name of this algorithm?
   6) Given that you have quick eyes, you can immediately determine, where to place the newly considered book into the stack of books on the right, what is its complexity of the overall arranging technique?

PART 3: ALL QUESTIONS ARE REQUIRED [30 marks]

Question 4: (10 marks)

a) Explain briefly what is meant by the Heap data-structure. (2 marks)

b) What is meant by Max Heap Property? Show how the heap will be built for a given array A[4,1,3,2,16,9,10,14,8,7] by drawing the heap in the first step and last step only after implementing the max-heapify function. (4 marks)

c) Find the right order for the following steps in order to implement the "Heap-Sort". (2 marks)
   1) Go to step .... Unless heap is ..... (complete the missing parts)
   2) Discard node n from heap
   3) Swap elements A[n] and A[1]
   4) Find maximum element A[1]
   5) Build Max-Heap from unordered array
   6) Run max-heapify

d) What is the complexity of the heap-sort? (2 marks)

Question 5: (10 marks)

Consider the Uniqueness Test Algorithm:
Check whether all the elements in a given array are distinct
- Input: An array A[0...n-1]
- Output: Return "true" if all the elements in A are distinct and "false" otherwise

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```
Algorithm UniqueElements ( A[0 … n-1])
for i ← 0 to n-2 do
    for j ← i+1 to n-1 do
        if A[i] = A[j] return false
```

a) Analyze the complexity of this algorithm. (4 marks)
b) Design an efficient algorithm to do the same job with lest complexity and drive the complexity (6 marks)

Question 6: (10 marks)

a) Fill the table below with the Big-O notation and the order (which is a number from 1 to 5, where 1 represents the asymptotically smallest and 5 represents the asymptotically largest): (Total 4 marks: 0.5 mark per each cell)

Expression	Big-O	Order
log n /4 + 2 n^2		
5000 n^2 log 4n		
70 n + 8000 n log 4		
50 n x 50 log n		

b) Use the appropriate method to find the O notation for the following recurrence relations.
[Note: your answer must show the steps that lead to your final answer]
1) $T(n) = 4T(n/2) + n^2$ (2 marks)
2) $T(n) = T(n-1) + n$ for $n > 0$ with $T(0) = 0$ (2 marks)

c) Find the complexity of the following blocks of code, where A, B are arrays, n is the input size: [Note: your answer must show the steps that lead to your final answer]

```
for i = 1 to n Do
        A[i] = 1 / i
endfor
SelectSort(A, n)
for i = 1 to n Do
        B[i] = i²
endfor
MergeSort(B, n)
```
(2 marks)

End of Questions

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